33 The arrow shows the path of an α-particle travelling towards the centre of a gold nucleus.



Which description of the path of the α-particle after striking the gold nucleus is correct?

A The α-particle is deflected through a very small angle towards the top of the page.

B The α-particle is deflected back along its initial path.

C The α-particle carries on in a straight line with **no** deflection.

D The α-particle is captured by the nucleus and goes into orbit around it.